University of Rajshahi

Department of Computer Science and Engineering

CSE4261 Neural Networks and Deep Learning

Assignment-11

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Problem - 1 : Comparing three face verifiers which are trained using Binary Cross-Entropy loss, contrastive loss and triplet loss, receptively.

As shown in Figure 1 and Figure 2, the images what are reconstruct using the BCE loss function

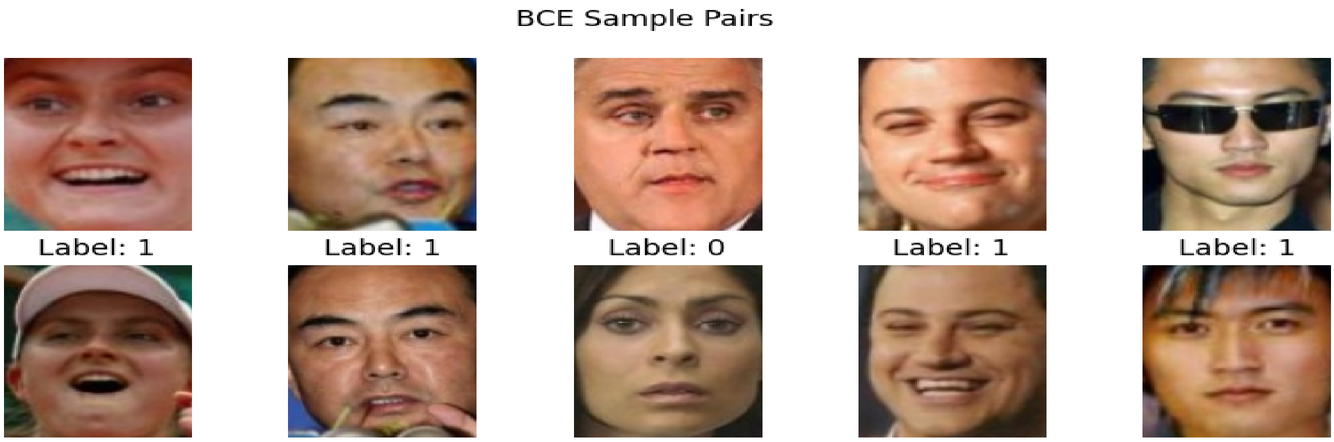

Figure 1: Output of the images using the BCE loss

Figure 2: Output of the training loss using the BCE loss

As shown in Figure 3 and Figure 4, the images what are reconstruct using the Contrastive loss function

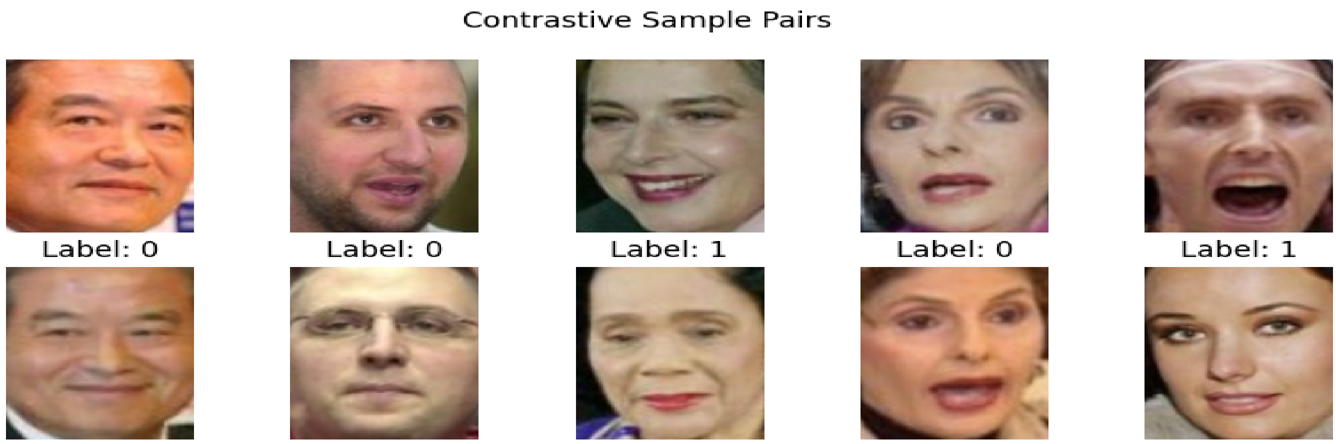

Figure 3: Output of the images using the Contrastive loss

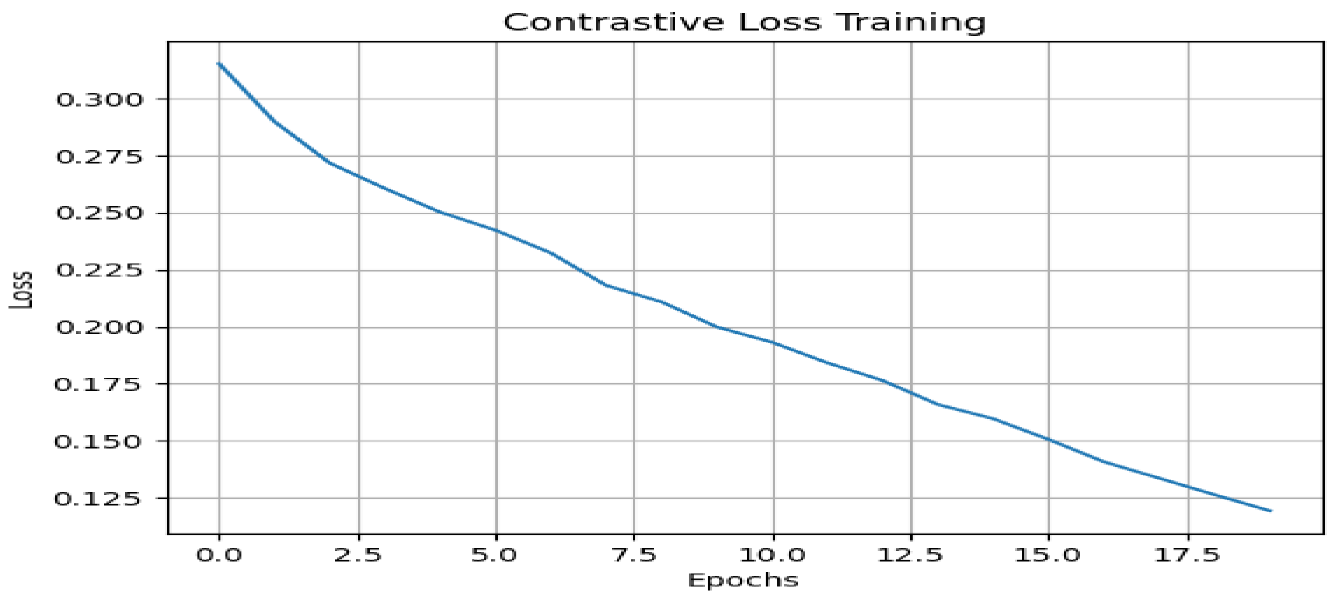

Figure 4: Output of the training loss using the Contrastive loss

As shown in Figure 5 and Figure 6, the images what are reconstruct using the Triplet loss function

Figure 5: Output of the images using the Triplet loss

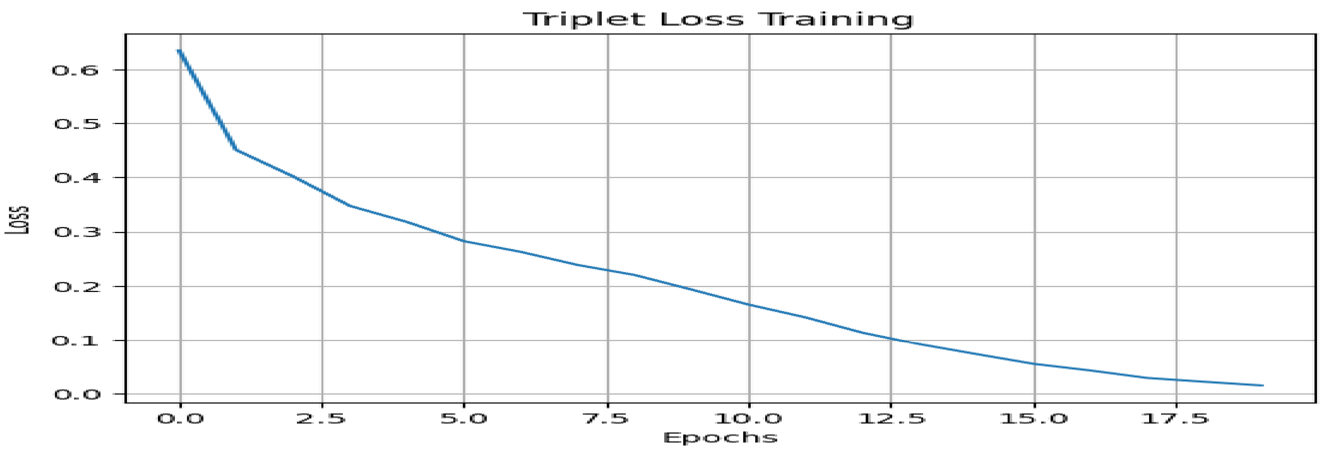

Figure 6: Output of the training loss using the Triplet loss

As shown in Figure 7, we can see that the loss of BCE loss is decreased too low. On the other hand, the Contrastive loss decreases very rapidly more than the BCE and Triplet loss. But, after increasing the epochs, the triplet loss is decreased more than the BCE loss and the Contrastive loss. After comparing all those three losses, we can see that the triplet loss is more beneficial for face verification than the BCE loss and contrastive loss.

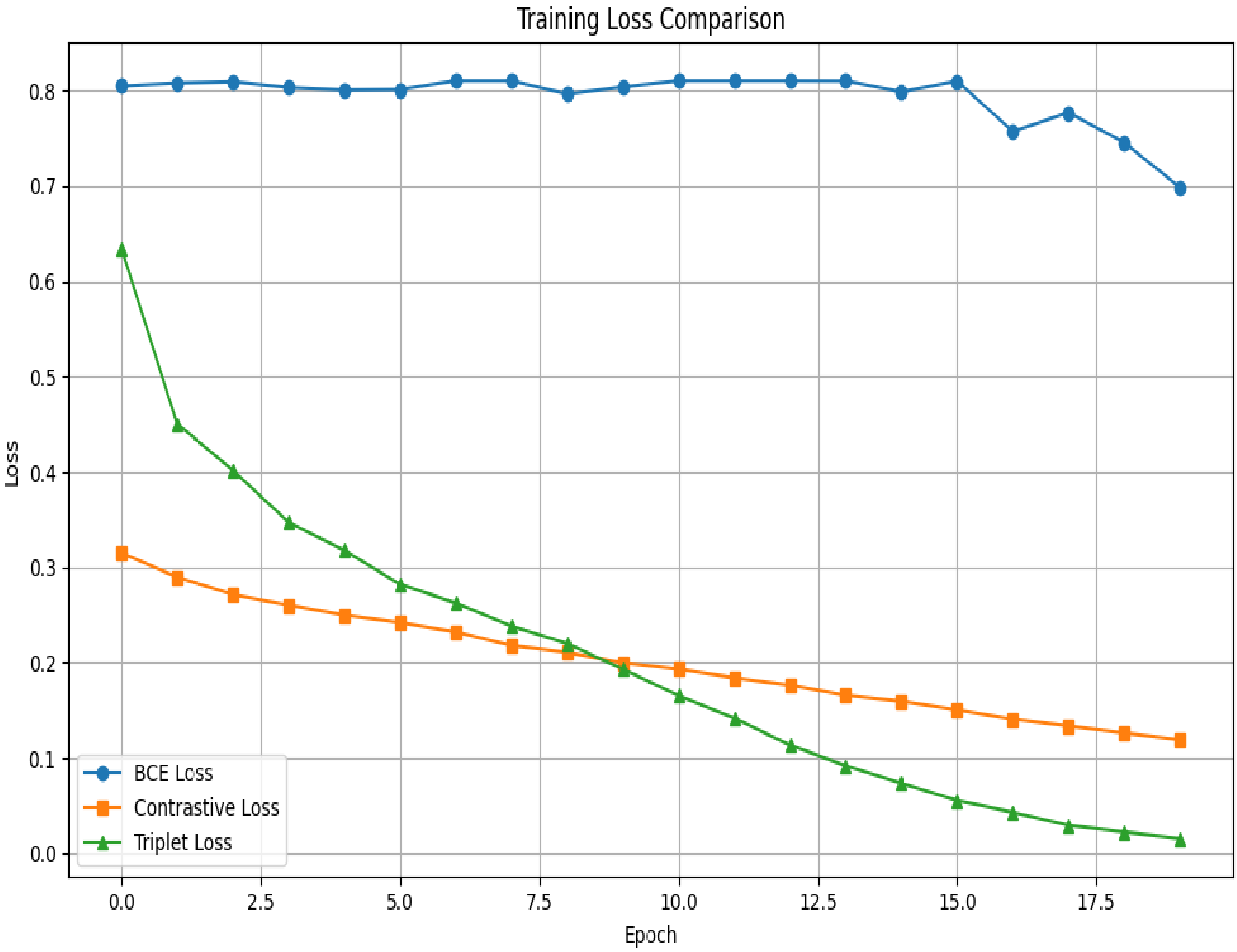

Figure 7: Output of BCE, Contrastive and Tiplet loss

Problem - 2: Comparing effect of Binary Cross-Entropy (BCE) loss and Mean Squared Error (MSE) as a reconstruction loss during training of a Variational Autoencoder (VAE).

As shown in Figure 8, the images what are reconstruct using the BCE loss function

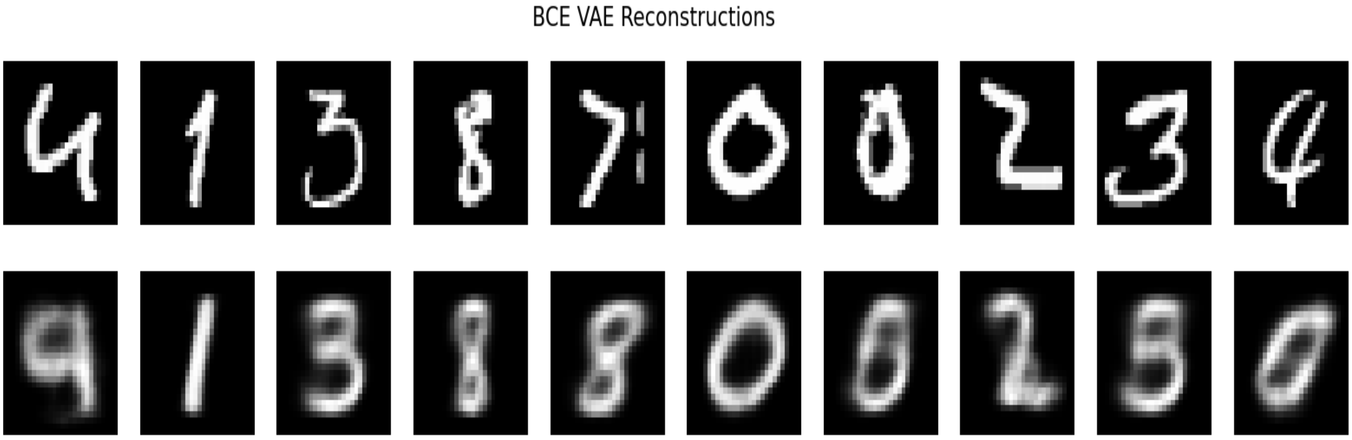

Figure 8: Output of the reconstructed images using the BCE loss

As shown in Figure 9, the images what are reconstruct using the MSE loss function

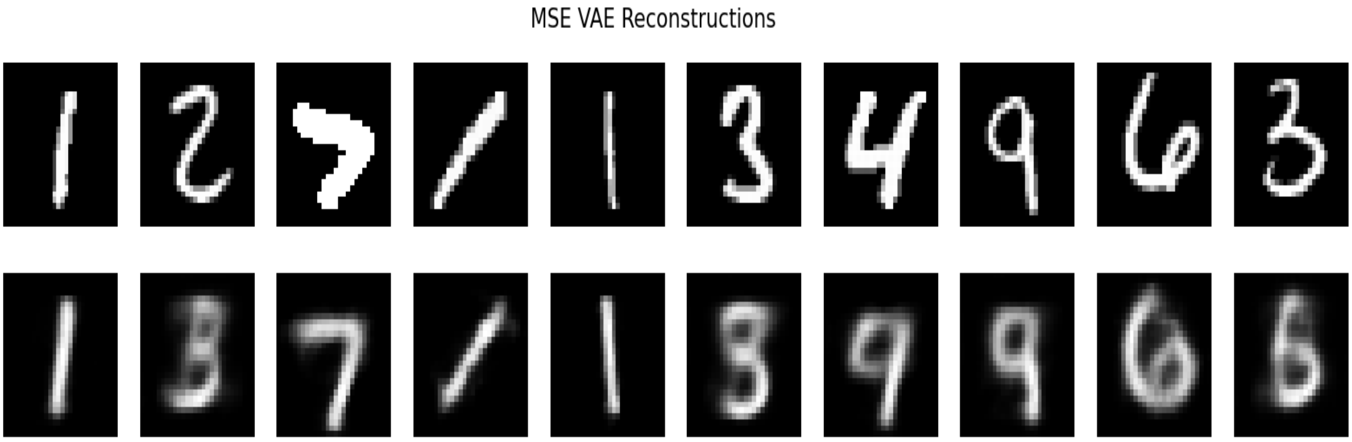

Figure 9: Output of the reconstructed images using the MSE loss

Assignment Github Link : Assignment